Ch 5 Review Questions Answers

1. The logical structure in which one instruction occurs after another with no branching is a _____.

    a. sequence

    b. selection

    c. loop

    d. case

    Answer: a

    Feedback:

    The logical structure with no branching is a sequence. Selections, loops, and cases all involve branching.

2. Which of the following is typically used in a flowchart to indicate a decision?

    a. square

    b. rectangle

    c. oval

    d. diamond

    Answer: d

    Feedback:

    Flowcharts typically use diamonds to represent decisions. Rectangles represent processing steps. Squares and ovals are not usually used.

3. Which of the following is not a type of `if` statement?

    a. single-alternative

    b. dual-alternative

    c. reverse

    d. nested

    Answer: c

Feedback:

"Reverse" is not a term used to describe any `if` statements; `if` statements can be single-alternative or dual-alternative, and either of those types can be nested.

4. A decision is based on a(n) _____ value.

    a. Boolean

    b. absolute

    c. definitive

    d. convoluted

    Answer: a

    Feedback:

    All decisions are based on a Boolean value: `true` or `false`.

5. In Java, the value of `(14 > 7)` is _____.

    a. `true`

    b. `false`

    c. 14

    d. 7

    Answer: a

    Feedback:

    The value of `14 > 7` (fourteen greater than seven) is `true`.

6. Assuming the variable `score` has been assigned the value 9, which of the following does not display *XXX*?

    a. `if(score <= 9) System.out.println("XXX");`

    b. `if(score > 9); System.out.println("XXX");`

    c. `if(score > 0); System.out.println("XXX");`

    d. `if(score == 5) System.out.println("XXX");`

    Answer: d

    Feedback:

When `score <= 9`, *XXX* is displayed because `score` is equal to 9. The value of `score` is not greater than 9, and it is greater than 0, but in each of those code segments, the first semicolon makes the output statement an independent statement, so both display *XXX*. The only code that does not produce output is when `score` is compared to 5.

7. What is the output of the following code segment?

```
t = 10;
if(t > 7)
{
   System.out.print("AAA");
   System.out.print("BBB");
}
```

   a. *AAA*

   b. *BBB*

   c. *AAABBB*

   d. nothing

   Answer: c

   Feedback:

   Because the value of t (10) is greater than 7, both *AAA* and *BBB* are output.

8. What is the output of the following code segment?

```
t = 0;
if(t > 7)
   System.out.print("AAA");
   System.out.print("BBB");
```

   a. *AAA*

   b. *BBB*

   c. *AAABBB*

   d. nothing

   Answer: b

Feedback:

Because the value of `t` (0) is not greater than 7, *AAA* is not output. However, the statement that displays *BBB* is a separate, stand-alone statement, not blocked as part of the decision.

9. What is the output of the following code segment?

```
t = 7;

if(t > 7)

{

   System.out.print("AAA");

   System.out.print("BBB");

}
```

   a. *AAA*

   b. *BBB*

   c. *AAABBB*

   d. nothing

   Answer: d

   Feedback:

   Because the value of `t` (7) is not greater than 7, neither *AAA* nor *BBB* is output.

10. When you code an `if` statement within another `if` statement, the statements are _____.

   a. notched

   b. nested

   c. nestled

   d. sheltered

   Answer: b

   Feedback:

   When an `if` statement is coded within another, the statement are nested.

11. The operator that combines two conditions into a single Boolean value that is `true` only when

both of the conditions are true is _____.

a. `$$`

b. `&&`

c. `||`

d. `!!`

Answer: b

Feedback:

The operator that combines two conditions into a single Boolean value that is `true` only when both of the conditions are true is `&&`. The `||` operator combines conditions and is `true` when either operand is `true`. `$$` is not a Java operator. `!!` is two NOT operators.

12. The operator that combines two conditions into a single Boolean value that is true when at least one of the conditions is true is _____.

a. `||`

b. `!!`

c. `$$`

d. `&&`

Answer: a

Feedback:

The operator that combines two conditions into a single Boolean value that is `true` when at least one of the conditions is true is `||`. The `&&` operator combines conditions and is `true` when both operands are `true`. `$$` is not a Java operator. `!!` is two NOT operators.

13. Assuming a variable `f` has been initialized to 5, which of the following statements sets `g` to 0?

a. `if(f > 6 || f != 5) g = 0;`

b. `if(f < 3 || f > 9) g = 0;`

c. `if(f >= 0 || f < 2) g = 0;`

d. `if(f == 0 || f == 2) g = 0;`

Answer: c

Feedback:

The statement that sets g to 0 is `if(f >= 0 || f < 2) g = 0;`. As soon as `f >= 0` is determined to be `true`, the entire `if` condition is `true`. In each of the other answer choices, neither part of the OR expression is `true`.

14. Which of the following has the lowest precedence?

    a. `<`

    b. `==`

    c. `&&`

    d. `||`

    Answer: d

    Feedback:

    The `||` operator has the lowest precedence.

15. Which of the following statements correctly outputs the names of voters who live in district 6 and voters who live in district 7?

    a.
    ```
    if(district == 6 || 7)
       System.out.println("Name is " + name);
    ```

    b.
    ```
    if(district == 6 || district == 7)
       System.out.println("Name is " + name);
    ```

    c.
    ```
    if(district = 6 && district == 7)
       System.out.println("Name is " + name);
    ```

    d.
    ```
    if(district != 6 && district != 7)
       System.out.println("Name is " + name);
    ```

    Answer: b

    Feedback:

    Determining if district == 6 or district == 7 selects the right voters.

16. Which of the following displays *Error* when a student ID is less than 1,000 or more than 9999?

a.
```
if(stuId < 1000) if(stuId > 9999)
    System.out.println("Error");
```

b.
```
if(stuId < 1000 && stuId > 9999)
    System.out.println("Error");
```

c.
```
if(stuId < 1000)
    System.out.println("Error");
else
    if(stuId > 9999)
      System.out.println("Error");
```

d.
```
 if(stuId < 1000 || > 9999)
    System.out.println("Error");
```

Answer: c

Feedback:

The code that displays the message when `stuId < 1000` is `true` and, otherwise, displays the message when `stuId > 9999` is the correct one. The code with two `if`s never displays anything. The code with the `&&` operator never displays anything. The code with the `||` operator would work if there were a complete Boolean expression on both sides of the operator.

17. You can use the _____ statement to terminate a switch statement.

a. `switch`

b. `break`

c. `case`

d. `end`

Answer: b

Feedback:

The `break` statement terminates a `switch` statement.

18. Which of the following cannot be the argument tested in a `switch` statement?

    a. `int`

    b. `char`

    c. `double`

    d. `String`

    Answer: c

    Feedback:

    A `double` cannot be used as the argument in a `switch` statement.

19. Assuming a variable `w` has been assigned the value 15, what does the following statement do?

```
w == 15 ? x = 2 : x = 0;
```

    a. assigns 15 to `w`

    b. assigns 2 to `x`

    c. assigns 0 to `x`

    d. nothing

    Answer: b

    Feedback:

    The statement compares `w` and 15, and if the expression is `true`, it assigns 2 to `x`.

20. Assuming a variable `y` has been assigned the value 6, the value of `!(y < 7)` is _____.

    a. 6

    b. 7

    c. `true`

    d. `false`

    Answer: d

    Feedback:

    The value of `y` is 6, so `y < 7` is `true`, so NOT `y < 7` is `false`.